

Artificial Intelligence (CS-203)

Objectives:

- K Nearest Neighbor Classifier
 - K-means Clustering
-

Classification:

The Classification algorithm is a Supervised Learning technique that is used to identify the category of new observations on the basis of training data. In Classification, a program learns from the given dataset or observations and then classifies new observation into a number of classes or groups. Such as, Yes or No, 0 or 1, Spam or Not Spam, cat or dog, etc. Classes can be called as targets/labels or categories.

The algorithm which implements the classification on a dataset is known as a classifier. There are two types of Classifications:

- **Binary Classifier:** If the classification problem has only two possible outcomes, then it is called as Binary Classifier.
Examples: YES or NO, MALE or FEMALE, SPAM or NOT SPAM, CAT or DOG, etc.
- **Multi-class Classifier:** If a classification problem has more than two outcomes, then it is called as Multi-class Classifier.
Example: Classifications of types of crops, Classification of types of music etc.

K Nearest Neighbor Classifier

K-Nearest Neighbor is one of the simplest Machine Learning algorithms based on Supervised Learning technique. It stores all the available data and classifies a new data point based on the similarity using following steps.

Step-1: Select the number K of the neighbors

Step-2: Calculate the Euclidean distance of K number of neighbors

Step-3: Take the K nearest neighbors as per the calculated Euclidean distance.

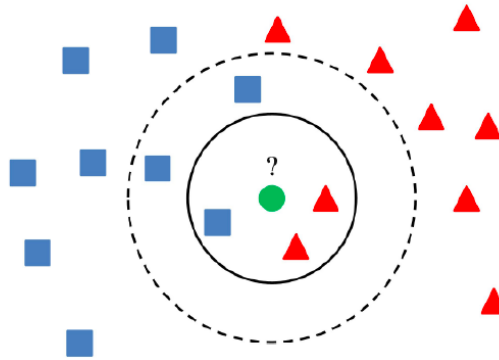
Step-4: Among these k neighbors, count the number of the data points in each category.

Step-5: Assign the new data points to that category for which the number of the neighbor is maximum.

The main idea behind K-NN is to find the K nearest data points to a given data point and then predict the label of the given data point based on the labels of its K nearest neighbors.

Find distance of test point with all training points

What would be class of green point if we consider K=3? And K=5?

**Example:**

To solve the numerical example on the K-nearest neighbor i.e. KNN classification algorithm, we will use the following dataset.

Point	Coordinates	Class Label
A1	(2,10)	C2
A2	(2, 6)	C1
A3	(11,11)	C3
A4	(6, 9)	C2
A5	(6, 5)	C1
A6	(1, 2)	C1
A7	(5, 10)	C2
A8	(4, 9)	C2
A9	(10, 12)	C3
A10	(7, 5)	C1
A11	(9, 11)	C3
A12	(4, 6)	C1
A13	(3, 10)	C2
A14	(3, 8)	C2
A15	(6, 11)	C2

In the above dataset, we have fifteen data points with three class labels. Now, suppose that we have to find the class label of the point P = (5, 7).

Point	Coordinates	Distance from P (5, 7)
A1	(2, 10)	4.24
A2	(2, 6)	3.16
A3	(11, 11)	7.21
A4	(6, 9)	2.23
A5	(6, 5)	2.23
A6	(1, 2)	6.40
A7	(5, 10)	3.0
A8	(4, 9)	2.23
A9	(10, 12)	7.07
A10	(7, 5)	2.82
A11	(9, 11)	5.65
A12	(4, 6)	1.41
A13	(3, 10)	3.60
A15	(3, 8)	2.23
A15	(6, 11)	4.12

Sorted Points according to distances are given below.

Point	Coordinates	Distance from P (5, 7)
A12	(4, 6)	1.41
A4	(6, 9)	2.23
A5	(6, 5)	2.23
A8	(4, 9)	2.23
A15	(3, 8)	2.23

A10	(7, 5)	2.82
A7	(5, 10)	3
A2	(2, 6)	3.16
A13	(3, 10)	3.6
A15	(6, 11)	4.12
A1	(2, 10)	4.24
A11	(9, 11)	5.65
A6	(1, 2)	6.4
A9	(10, 12)	7.07
A3	(11, 11)	7.21

For this, we will first specify the number of nearest neighbors i.e. k . Let us take k to be 3. Now, we will find the distance of P to each data point in the dataset. For this KNN classification numerical example, we will use the euclidean distance metric. The following table shows the euclidean distance of P to each data point in the dataset.

Distance of each point for the new point

After finding the distance of each point in the dataset to P , we will sort the above points according to their distance from P (5, 7). After sorting, we get the following table.

As we have taken $k=3$, we will now consider the class labels of three points in the dataset nearest to point P to classify P . In the above table, A12, A4, and A5 are the closest 3 neighbors of point P . Hence, we will use the class labels of points A12, A4, and A5 to decide the class label for P .

Now, point A12, A4, and A5 have the class labels C1, C2, and C1 respectively. Among these points, the majority class label is C1. Therefore, we will specify the class label of point $P = (5, 7)$ as C1. Hence, we have successfully used KNN classification to classify point P according to the given dataset.

By studying the above KNN classification numerical example, you can see that the algorithm is pretty straightforward and doesn't require any specific mathematical skills apart from distance calculation and majority selection.

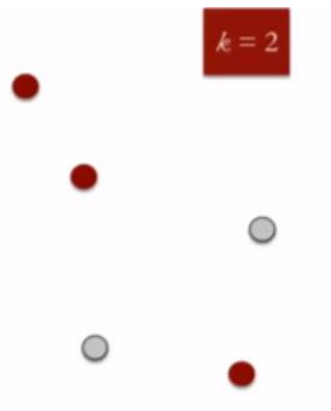
Clustering:

Clusters are collections of data based on similarity. Clustering is the task of dividing the data points into a number of groups such that data points in the same groups are more similar to other data points in the same group than those in other groups. There are many types of clustering algorithms among which one is K-Mean Clustering.

K-Means Clustering:

K means is an iterative clustering algorithm that aims to find local maxima in each iteration. This algorithm works in these 5 steps:

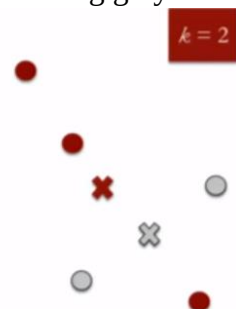
Step1: Specify the desired number of clusters K: Let us choose $k=2$ for these 5 data points in 2-D space.



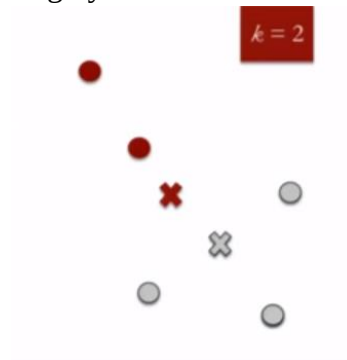
Step2: Randomly assign each data point to a cluster : Let's assign three points in cluster 1 shown using red color and two points in cluster 2 shown using grey color.



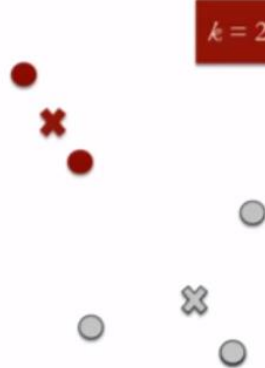
Step3: Compute cluster centroids : The centroid of data points in the red cluster is shown using red cross and those in grey cluster using grey cross.



Step4: Re-assign each point to the closest cluster centroid : Note that only the data point at the bottom is assigned to the red cluster even though its closer to the centroid of grey cluster. Thus, we assign that data point into grey cluster.



Step5: Now, re-computing the centroids for both the clusters. Repeat until there will be no further switching of data points between two clusters.



Example1:

Consider the following set of points

P1(1,3), P2(2,2), P3(3,1), P4(4,2), P5(8,1), P6(8,5), P7(9,2), P8(10,1)

Consider that clusters are initialized at : C1(2,1) and C2(9,2)

Apply distance formula :

$$P(x, y) = |x_2 - x_1| + |y_2 - y_1|$$

- (a) Draw a table in the format below showing which point belongs to which cluster (clearly show distances for each point from both clusters)?

	C ₁	C ₂
P _i	Distance of P _i from C ₁	Distance of P _i from C ₂
P _j	Distance of P _j from C ₁	Distance of P _j from C ₂

SOLUTION

Calculate distance using given formula as follows and fill table

$$d(P1, C1) = |3 - 1| + |1 - 2| = |2| + |1| = 3$$

$$d(P1, C2) = |9 - 1| + |2 - 3| = |8| + |1| = 9$$

$$d(P2, C1) = |2 - 2| + |2 - 1| = |0| + |1| = 1$$

$$d(P2, C2) = |9 - 2| + |2 - 2| = |7| + |0| = 7$$

Points	C1	C2
P1	3	9
P2	1	7
P3	1	7
P4	3	5
P5	6	2
P6	10	4
P7	8	0
P8	8	2

Now we can decide which points belong to which cluster based on distance

Cluster1: P1, P2, P3, P4

Cluster2: P5, P6, P7, P8

(b) Show the first iteration where the centroid center and shift from their default position C1(2,1) and C2(9,2) according to points calculated in part (a).

SOLUTION

Centroid1:

$$X = (1+2+3+4)/4$$

$$Y = (3+2+1+2)/4$$

New Centroid Center: (2.5, 2)

Centroid2:

$$X = (8+8+9+10)/4$$

$$Y = (1+5+2+1)/4$$

New Centroid Center: (8.75, 2.25)
